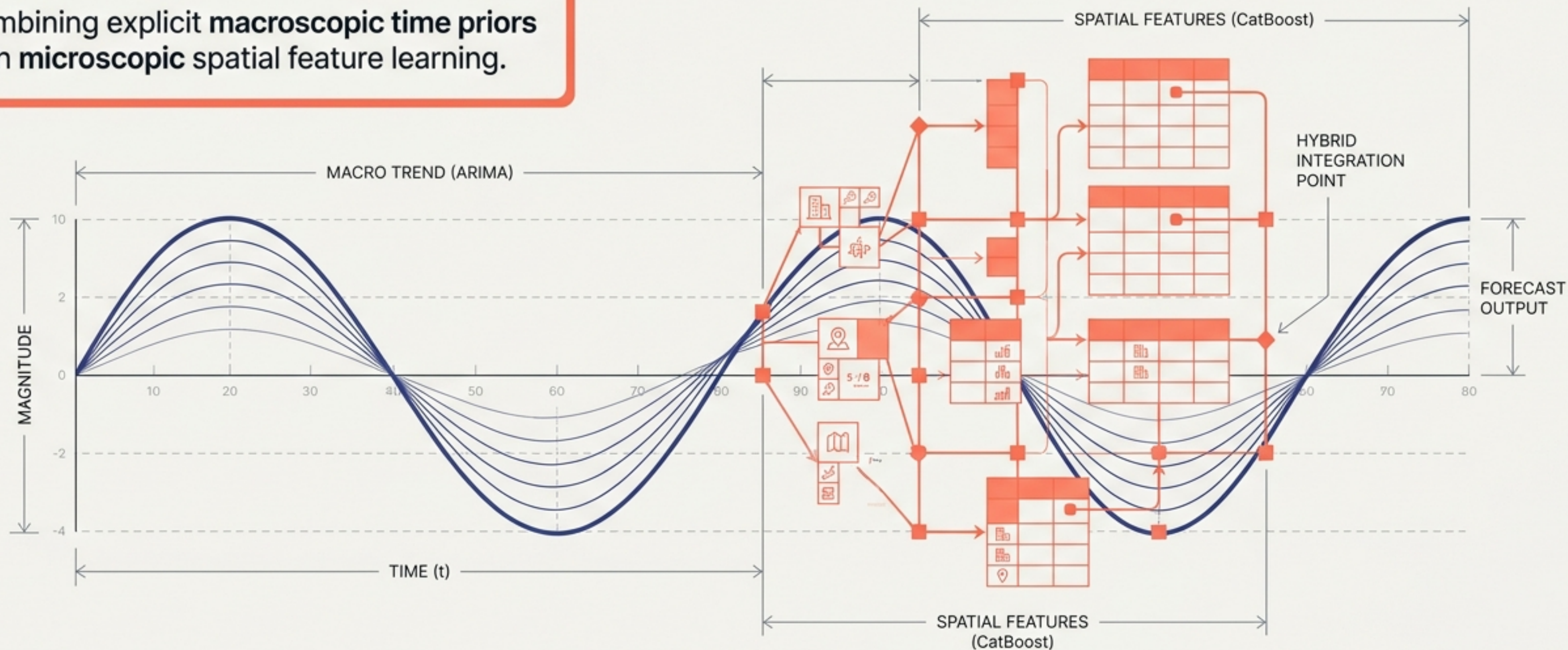


The V4 Forecasting Architecture

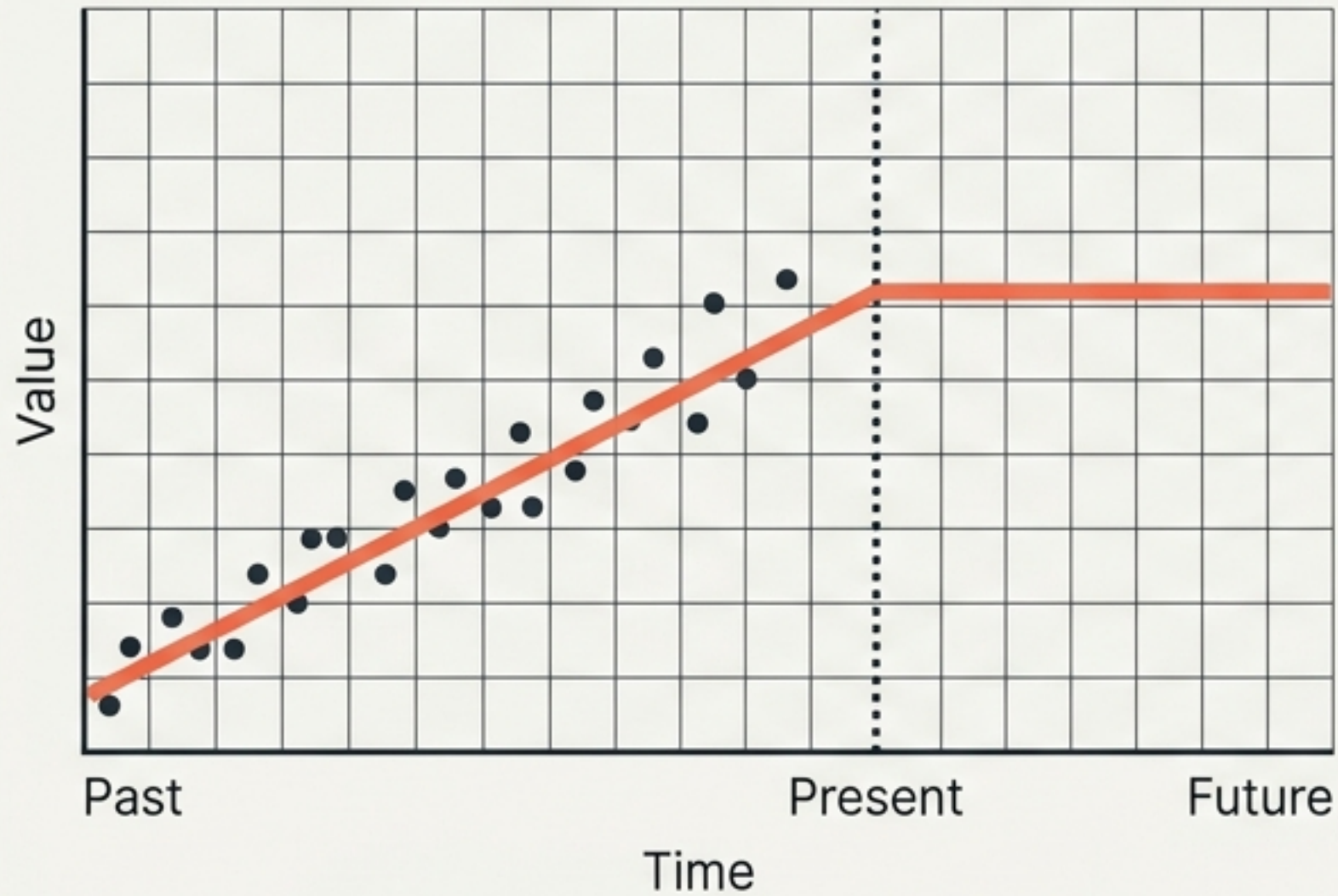
A Hybrid ARIMA-CatBoost Pipeline for Real Estate Pricing.

Combining explicit **macroscopic time priors** with **microscopic spatial feature learning**.



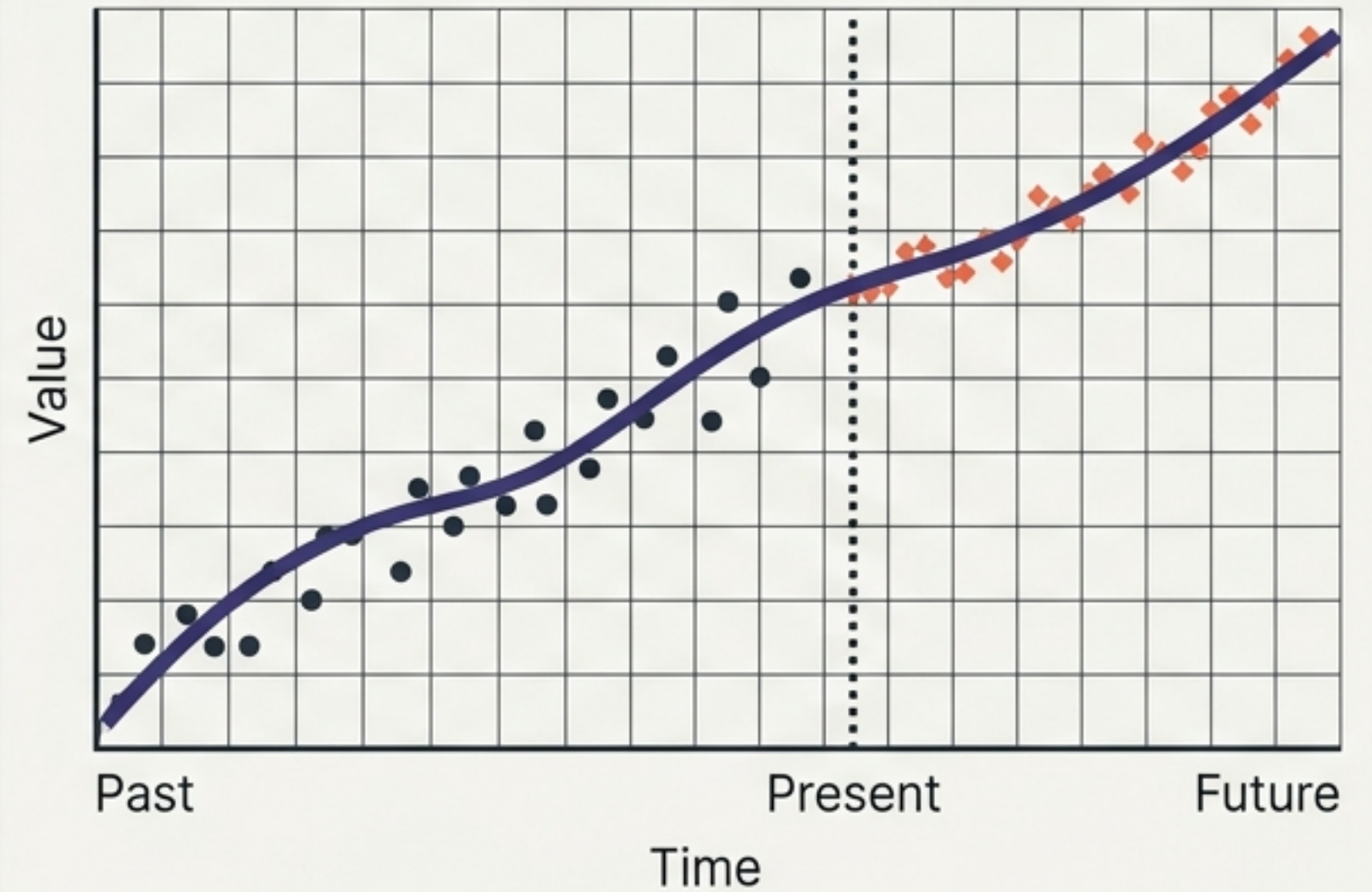
Trees Fail at Long-Horizon Autocorrelation

The Problem: Flatlining Trees



Tree models are exceptional tabular function approximators but lack natural long-horizon temporal memory.

The V4 Approach: Hybrid Extrapolation

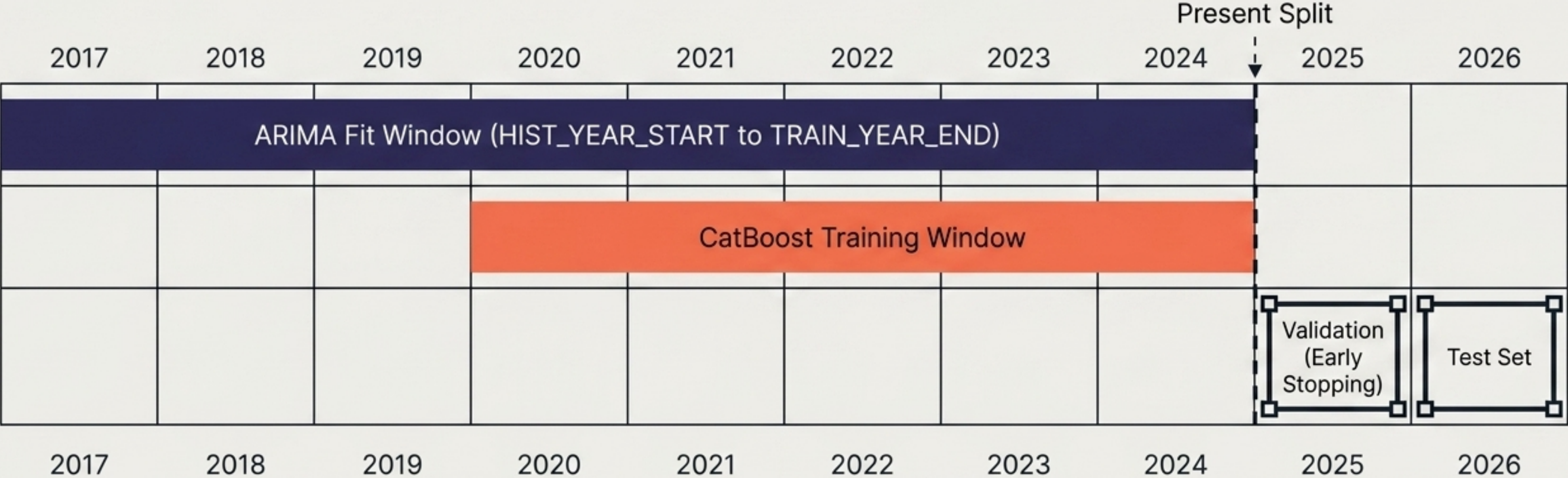


V4 solves this by explicitly separating the temporal drift of the market from the physical attributes of the individual property.

Dividing the Labor: Macro Time vs. Micro Space

ARIMA (The Macro Layer)		CatBoost (The Micro Layer)
Cheap, explicit time-series prior.	Model Role	Flexible tabular function approximator.
Market aggregates (Monthly mean log_resale_price).	Target Scope	Unit-level rows (Individual resale transactions).
Long-horizon autocorrelation, smoothing, and trends.	Features Learned	Spatial KDTree encodings, floor area, lease, location, amenities.
Summarizes the slowly moving market structure and historical volatility.	Function in v4	Maps the full physical feature vector to price, using ARIMA's output as macroeconomic context.

Preventing Future-State Leakage



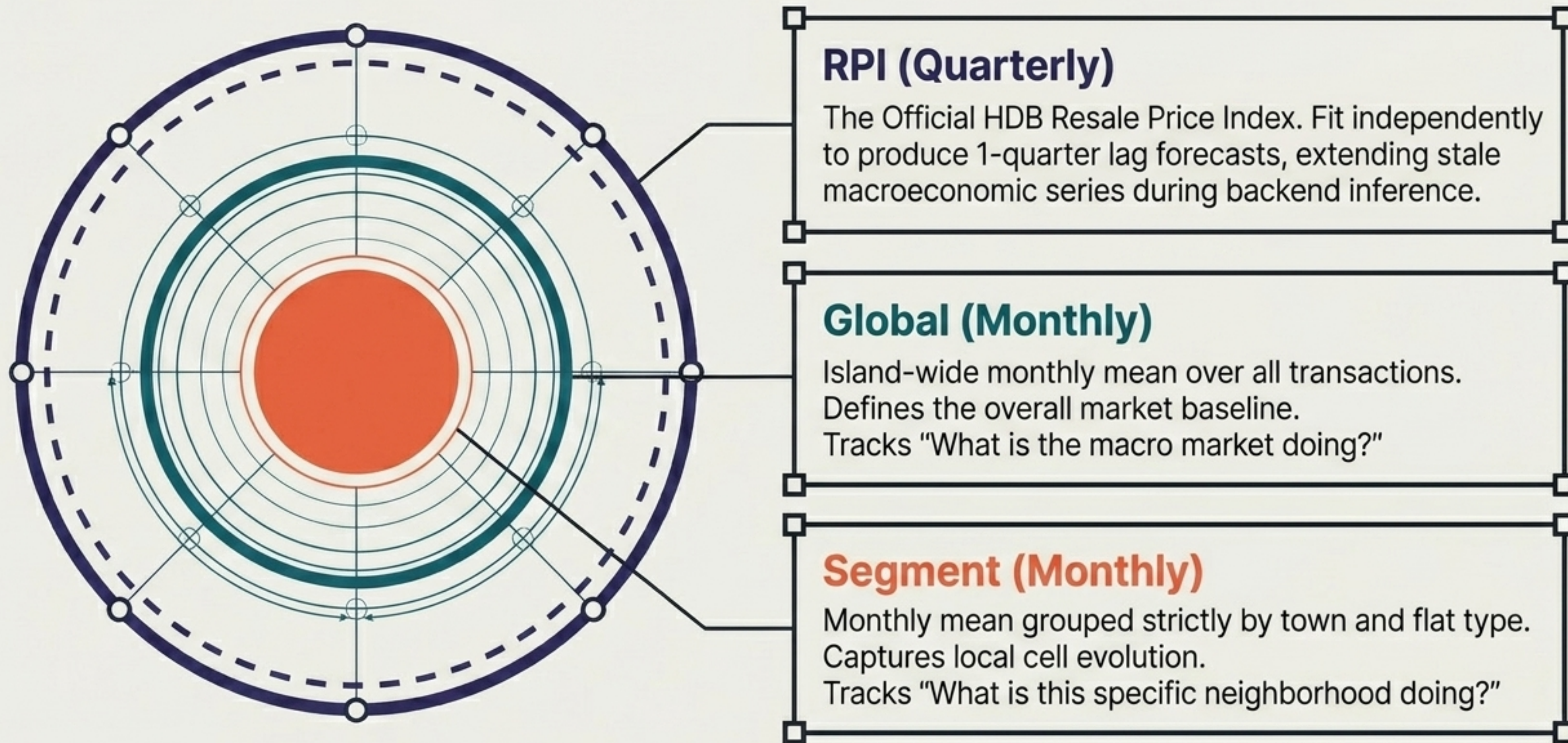
Strict Isolation

ARIMA only sees the broad historical envelope (2017-2024). It never peeks at validation or test years.

Static Risk

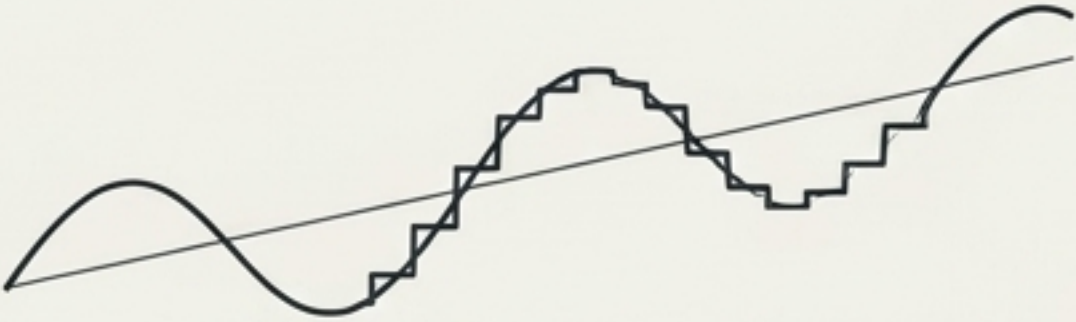
Volatility signals remain constant across train, validation, and test sets within a segment.

Three Tiers of Market Aggregation

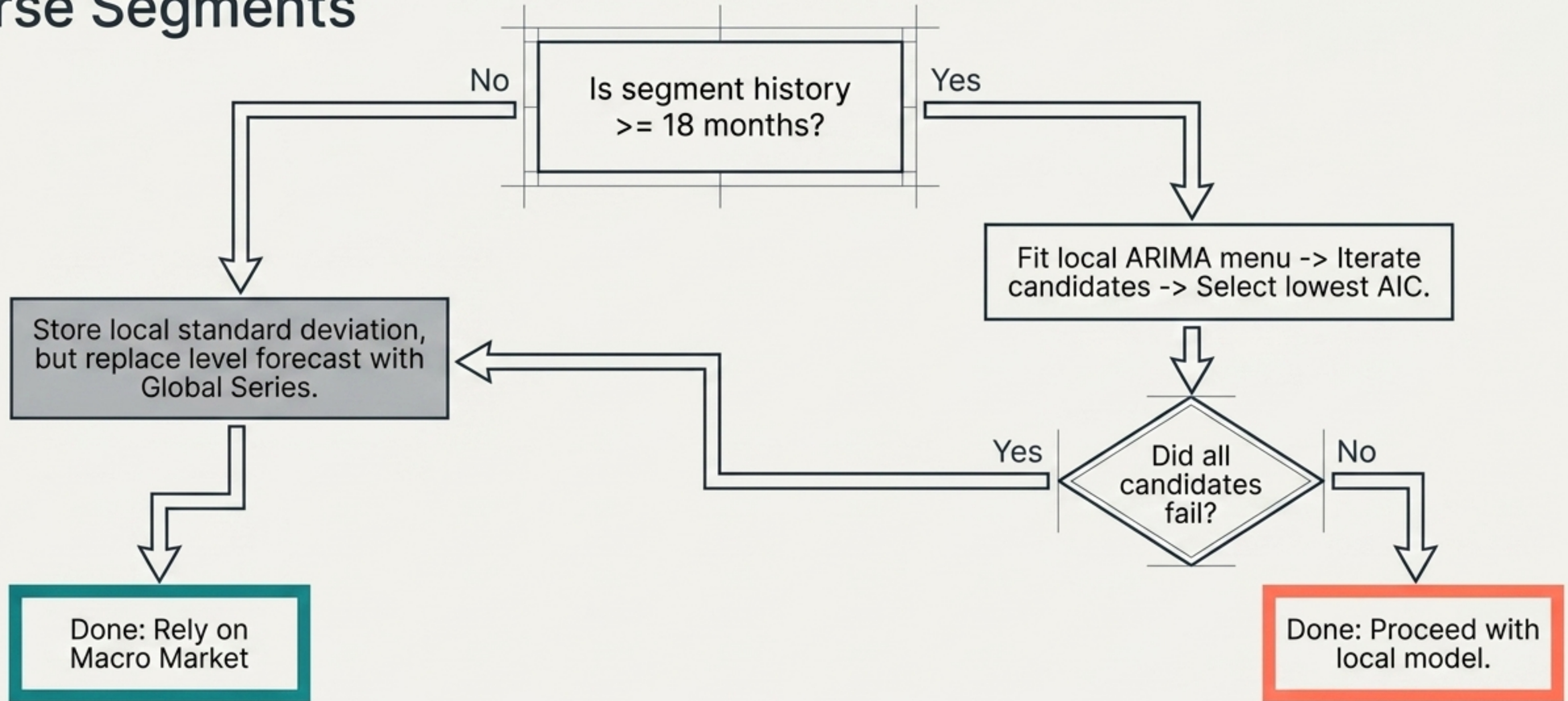


Encoding Market Memory

Integration (d=1 everywhere): Standard for non-stationary levels. We difference to target growth/cooling rather than absolute price.

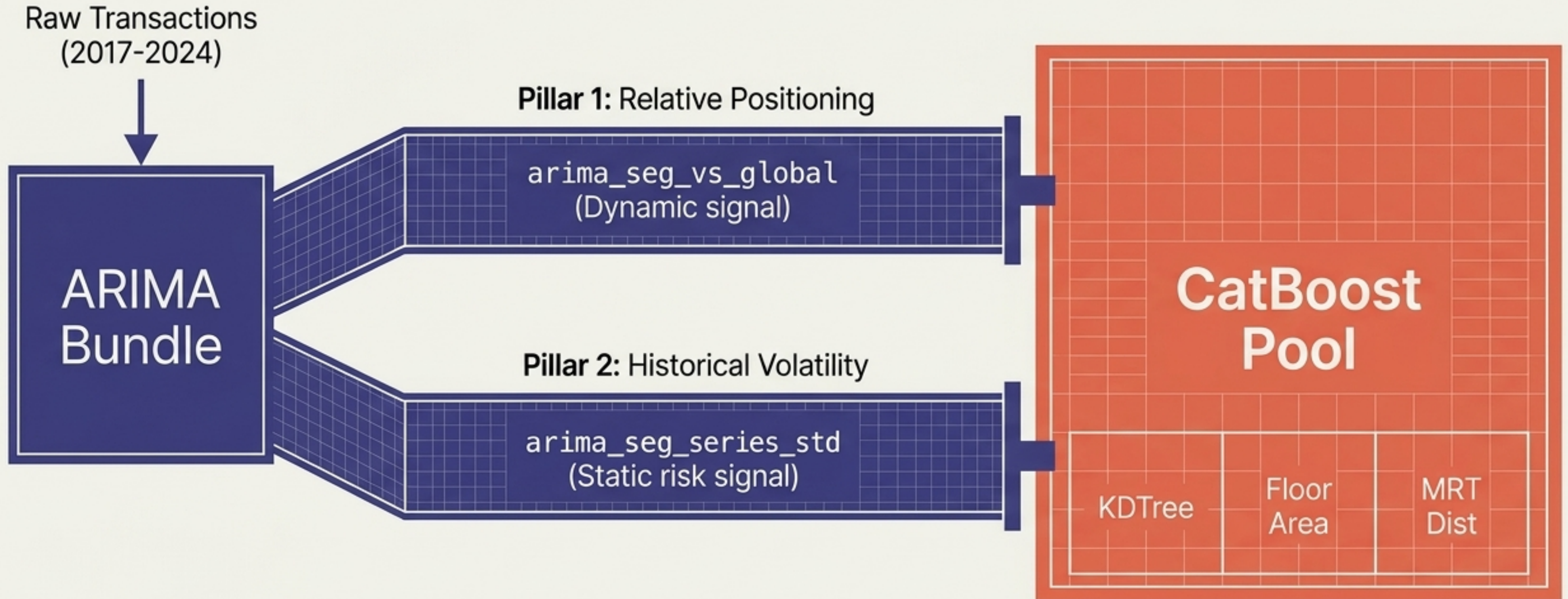
The Math	The Shape	The Intuition
$(0, 1, 1)$		Local Trend: Smooth, slowly drifting level with short-memory noise. Wins on smooth market aggregates.
$(1, 1, 0)$		Momentum: Mean-reversion in growth rates. Today's change depends explicitly on yesterday's change.
$(1, 1, 1)$		The Blend: A flexible, workhorse mix of persistence and correlated shocks.
$(2, 1, 1) / (2, 1, 2)$		Multi-step Cycles: Richer memory allowing for overshoots. Restricted to the Global series to prevent local overfitting.

Graceful Degradation for Sparse Segments



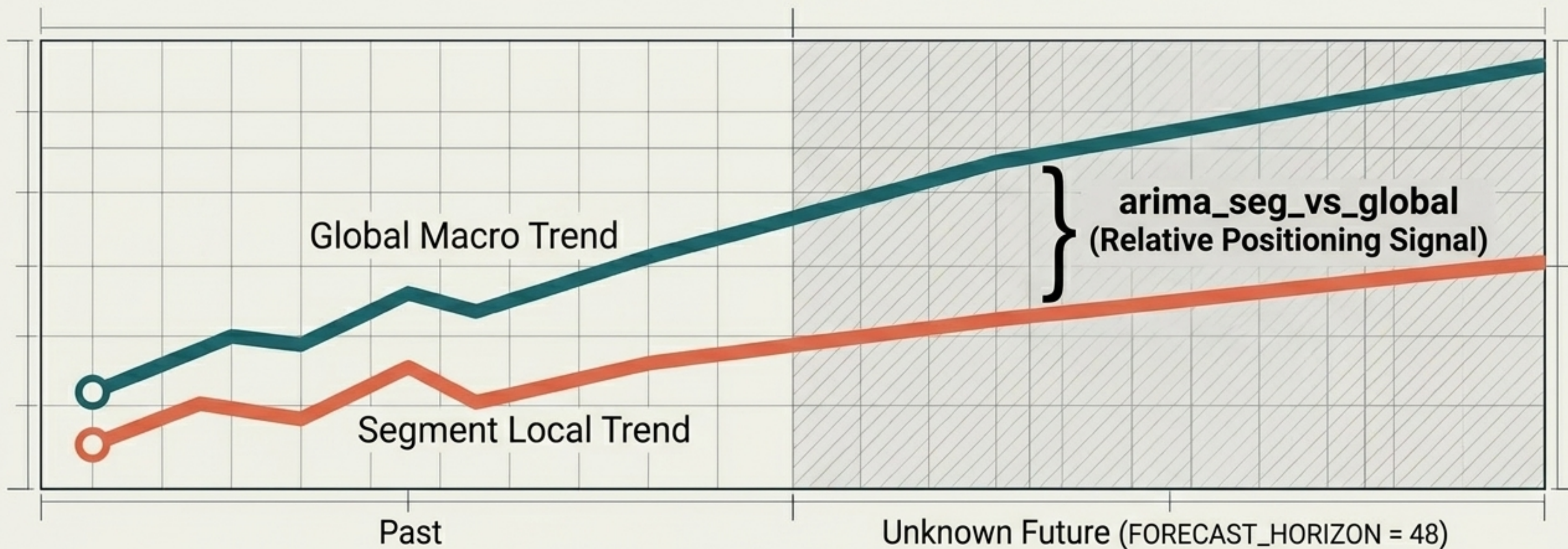
We avoid unstable, high-parameter fits on thin data. The system falls back to the global market level while safely retaining the local cell's historical volatility signature.

The Handoff: Distilling Time into Two Contextual Pillars



The Result: CatBoost does not ingest raw time series; it receives a strictly low-dimensional summary of temporal context.

Relative Positioning Hedges Against Future Drift



Absolute forecasts can inevitably drift or bias when extrapolating far into distant future months.

By feeding the tree model the difference between the segment and global forecasts, we safely rank hot vs. cold neighborhoods.

Even if the macroscopic baseline is fundamentally wrong, the cross-sectional ranking remains intact.

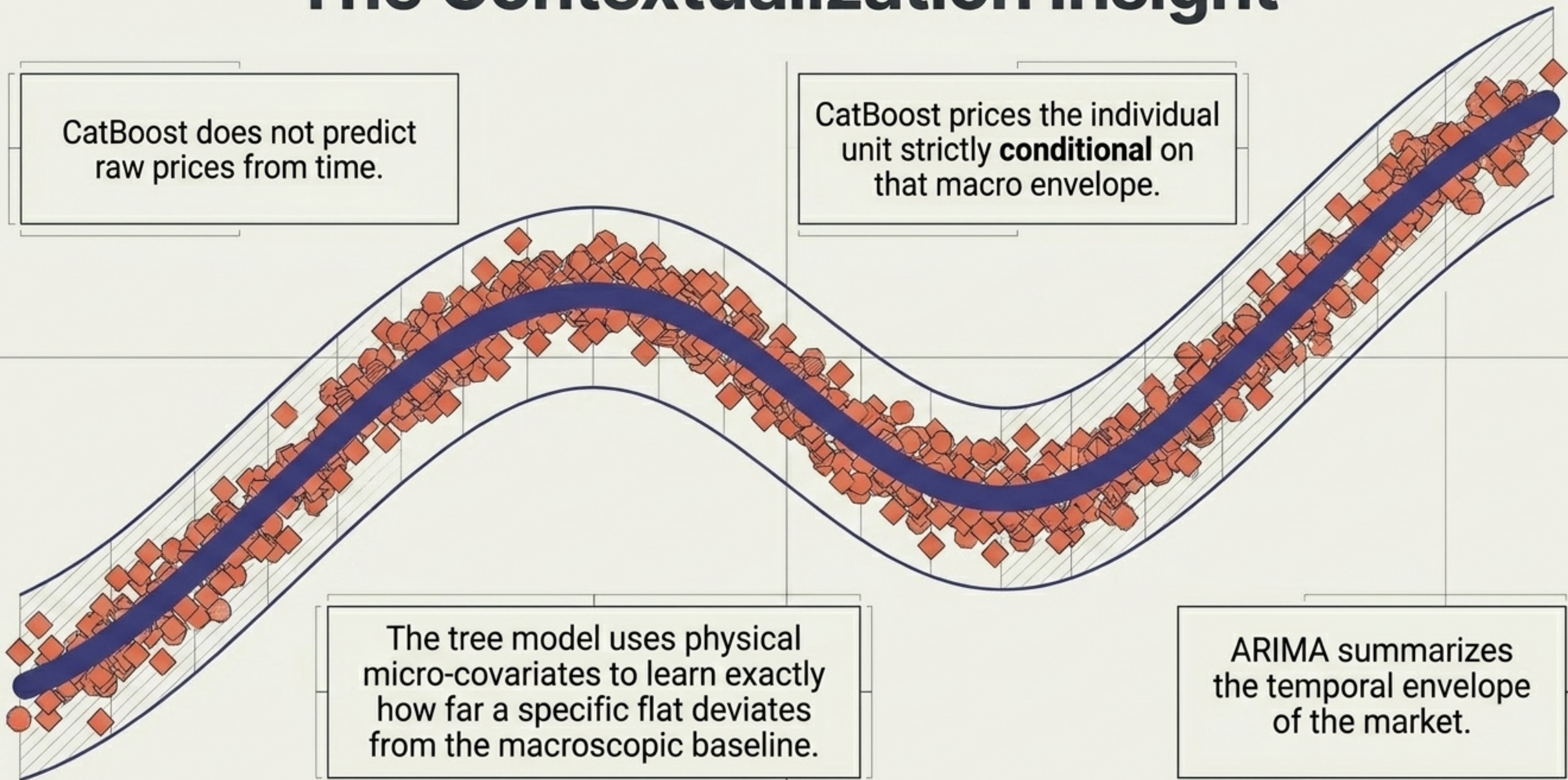
The Contextualization Insight

CatBoost does not predict raw prices from time.

CatBoost prices the individual unit strictly **conditional** on that macro envelope.

The tree model uses physical micro-covariates to learn exactly how far a specific flat deviates from the macroscopic baseline.

ARIMA summarizes the temporal envelope of the market.



Extending Macro Indicators at Inference

Training Phase



Official HDB
RPI CSV

Model trains on actual
hdb_rpi where published.

CatBoost
Feature Matrix

The official RPI strictly stops at the last published quarter. Queries for future dates (e.g., 2026) would hit stale data.

The RPI ARIMA branch exists exclusively to cleanly extend this macro series in production.

Inference Phase (2026+)



Stale or Missing
RPI Data



**RPI ARIMA
Fallback Mechanism**

CatBoost
Feature Matrix

This ensures backend inference resilience without altering the core feature set.

Architectural Trade-Offs

The Architectural Benefits



The Accepted Costs



The hybrid stack deliberately trades operational simplicity for explicit structural control. We accept the maintenance overhead of syncing two model states to gain absolute interpretability over the temporal baseline.

Reference Pipeline Performance

Validation Set (2025)	Test Set (2026 Forward-Year)	System Stability
MAE: ~26.2k MAPE: ~3.9%	MAE: ~29.6k MAPE: ~4.4%	$R^2 \sim 0.95+$
Used strictly for early stopping.	Absolute error tightly bounded despite forward-drift.	Maintained reliably on the unknown test set.

Operational Specs: Training executes rapidly (~39s for CatBoost after feature extraction). The target metric strictly utilizes expm1 to recover exact dollar values from the log target.

Conclusion: The minimal degradation from Validation to Test successfully proves the efficacy of the ARIMA relative positioning strategy.